

2021

Agras T10 Repair Training

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01 Product Introduction

01 Product Introduction



Product Name: Agras T10

- Brand-new design, including a quadrilateral folding structure and a **quick-release spray tank and flight battery** that makes replacement, installation, and storage easy.
- Updated Route Operation mode, which enables marking obstacles outside the field planning, enabling more intelligent flights and better automated operations.
- The **Spherical Perception Radar System** supports functions such as terrain following, obstacle sensing, and obstacle circumvention. With the **forward and backward FPV cameras** and bright spotlights, the system comprehensively ensures operational safety day and night in different weather.
- Spraying System: Equipped with a **10L spray tank, four sprinklers,** and a **2-channel electromagnetic flow meter** that provides even and accurate spraying, users can save liquid and reduce operating costs.
- With a protection rating of IP67 and with key components boasting three layers of protection, the T10 is corrosion-resistant, dustproof, and waterproof so that it can be washed directly with water.

01 Product Introduction—Product Comparison - Aircraft



Agras T10

- 10L detachable spray tank
- Spherical Perception Radar System
- 2-Channel Electromagnetic Flow Meter
- Single Point Liquid Level Gauge (Makes alert when tank is empty)
- Dual FPV Cameras Real-time Monitor
- IP67
- Smart Agriculture Cloud Platform
- 4 sprinklers



Agras T30

- 30L fixed spray tank
- Spherical Perception Radar System
- 2-Channel Electromagnetic Flow Meter
- Continuous Liquid Level Gauge (Real-time spray measurement and Smart Supply Reminder)
- Dual FPV Cameras Real-time Monitor
- IP67
- Smart Agriculture Cloud Platform
- 16 sprinklers

01 Product Introduction—Product Comparison - Remote Controller



Both T10 and T30 drones are used with the DJI Smart Controller 3.0, and compared with the T20 remote controller, the following features have been changed:

- Added a flip-up cover to optimize battery protection
- Uses DJI OcuSync 2.0, with a maximum transmission range of up to 5km
- Added a DJI Agras logo to the front side of the remote controller

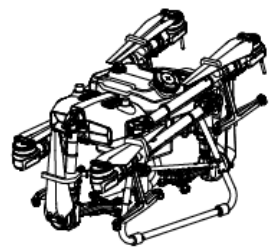
01 Product Introduction—Specifications

		Agras T10	Agras T30
	Hourly operating efficiency	100 acre	240 acre
Aircraft	Total Weight (Excluding Battery)	13 kg	26.3 kg
	Max Takeoff Weight	26.8 kg (at sea level)	66.5 kg (at sea level)
	Hovering Accuracy (With a strong GNSS signal)	D-RTK enabled: Horizontal: ± 10 cm, Vertical: ± 10 cm D-RTK disabled: Horizontal: ± 0.6 m, Vertical: ± 0.3 m (Radar module enabled: ± 0.1 m)	D-RTK enabled: Horizontal: ± 10 cm, Vertical: ± 10 cm D-RTK disabled: Horizontal: ± 0.6 m, Vertical: ± 0.3 m (Radar module enabled: ± 0.1 m)
	Max Power Consumption	3700 W	13000 W
	Hovering Time	19 min (takeoff weight of 16.8 kg with a 9500 mAh battery) 8.7 min (takeoff weight of 26.8 kg with a 9500 mAh battery)	20.5 min (takeoff weight of 36.5 kg with a 29000 mAh battery) 7.8 min (takeoff weight of 66.5 kg with a 29000 mAh battery)
Airframe	Max Diagonal Wheelbase	1480 mm	2145 mm
	Dimensions	1958 mm×1833 mm×553mm (Arms and propellers unfolded) 1232 mm×1112 mm×553 mm (Arms unfolded and propellers folded) 600 mm×665 mm×580 mm (Arms folded)	2858 mm×2685 mm×790mm (Arms and propellers unfolded) 2030 mm×1866 mm×790 mm (Arms unfolded and propellers folded) 1170 mm×670mm×857 mm (Arms folded)
Propulsion System	Motor Stator Size	100×10 mm	100×18 mm
	Motor Max Power	2000 W/rotor	3600 W/rotor
	Propeller Diameter	33×9 inches	38×20 inches
	ESC Max Working Current (Continuous)	32 A	60 A

01 Product Introduction—Specifications

		Agras T10	Agras T30
Spraying System	Spray Tank Volume	Fully loaded 10 L	Fully loaded 30L
	Spray Tank Installation Method	Quick-release	Fixed
	Liquid Level Gauge	Single Point Liquid Level Gauge	Continuous Liquid Level Gauge (Real-time spray measurement and Smart Supply Reminder)
	Operating Payload	Fully loaded 10 kg	Fully loaded 30kg
	Nozzle Model	SX11001VS (Standard) SX110015VS (Optional) XR11002VS (Optional)	SX11001VS (Standard) SX110015VS (Optional) ZX-VK04 (Optional for Orchard Configuration)
	Quantity	4	16
	Max Spray Rate	SX11001VS: 1.8 L/min SX110015VS: 2.4 L/min XR11002VS: 3 L/min	SX11001VS: 7.2 L/min SX110015VS: 8 L/min ZX-VK4: 3.6 L/min
	Droplet Size	SX11001VS: 130 - 250 μm SX110015VS: 170 - 265 μm XR11002VS: 190 - 300 μm (Subject to operating environment and spray rate)	SX11001VS: 130 - 250 μm SX110015VS: 170 - 265 μm ZX-VK4: 110 - 135 μm (Subject to operating environment and spray rate)
	Spray Width	3-5.5 m (4 nozzles, at a height of 1.5 - 3 m above crops)	4-9 m (12 nozzles, at a height of 1.5 - 3 m above crops)
	Pump Model	Diaphragm pump	Plunger pump
Battery	Pump operating voltage	15 V	60 V
	Max Pump Rate	1.5 L/min ×1	4 L/min ×1
	Model	BAX501-9500mAh-51.8V	BAX501-29000mAh-51.8V
	Weight	Approx. 3.8 kg	Approx. 10.2 kg
Remote Controller	Capacity	9500 mAh	29000 mAh
		51.8 V	51.8 V
	Model	RM500-ENT	RM500-ENT
	Operating Frequency	2.4000 GHz to 2.4835 GHz 5.725 GHz to 5.850 GHz	2.4000 GHz to 2.4835 GHz 5.725 GHz to 5.850 GHz
	Effective Transmission Distance (Unobstructed, free of interference)	SRRC: 5 km MIC/KCC/CE: 4 km FCC: 7 km (Operation height: 2.5 m)	SRRC: 5 km MIC/KCC/CE: 4 km FCC: 7 km (Operation height: 2.5 m)

01 Product Introduction—In the Box



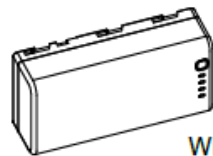
x 1

Aircraft (inc. Spraying System and Propeller Holders)



x 1

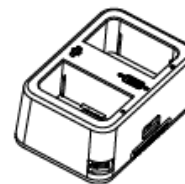
Remote Controller



x 1

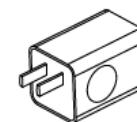
Remote Controller Intelligent Battery

WB37



x 1

Intelligent Battery Charging Hub



x 1

USB Charger



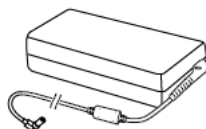
x 1

USB-C Cable



x 1

Remote Controller Lanyard



x 1

Power Adapter



x 1

Charging Hub AC Power Cable



x 1

4G Dongle



Tools

3.0 mm Hex Key
4.0 mm Hex Key
Double-Ended Wrench



Spare Screws, Nuts, and Other Accessories

Hose
Hose Nuts A
Hose Nuts B
Frame Arm Pins
Propeller Washers
M3×8 Screws, M3×10 Screws,
M4×10 Screws, M5×10 Screws



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02 Material Identification

02 Material Identification—Identify the Module



Omnidirectional Digital Radar



Upward Radar

Spherical Perception Radar System includes:

Omnidirectional Digital Radar + Upward Radar

Supports functions such as terrain following, obstacles sensing, and obstacle circumvention, comprehensively ensuring operational safety day and night in different weather; additionally, it is dustproof and free of lighting interference.

02 Material Identification—Identify the Module



Cable Distribution Module



Aerial-Electronics System Module + Spraying Module + RF Module + Upward Radar +
Cable Distribution Board Module

The control module uses a fully enclosed structure for added durability. Three layers of protection over key components offer an overall rated water resistance of IP67 against pesticides, dust, fertilizers, and corrosion.

02 Material Identification—Identify the Module



FPV Module

Equipped with dual FPV cameras, the T10 provides clear front and rear views and enables you to check the flight status without turning. Furthermore, its bright spotlight doubles the drone's night vision capabilities, creating more possibilities for night operation.



Power Distribution Module

The power distribution module and the cable distribution module have fully enclosed potting protection..



RTK Antenna Module

The RTK module provides information such as position, direction, and speed of the aircraft for the flight controller system to ensure the stability of the aircraft navigation and the accuracy of operation.

02 Material Identification—Identify the Module



ESC Module

The ESC module is responsible for converting the throttle information into the motor speed information, controlling the motor rotation, reporting the motor speed to the flight controller, and providing functions such as overcurrent protection.



10010S Motor

The motor is the executive part of the aircraft, which drives the propellers to provide the upward propulsion.



Rear Indicator Module

The rear indicator shows the current operation status of the aircraft through the color of the LEDs.

02 Material Identification—Identify the Module



Delivery Pump Module



Electromagnetic Exhaust Valves

Spraying System: Equipped with a diaphragm pump, four sprinklers, a 2-channel electromagnetic flow meter, and four electromagnetic exhaust valves, the spraying system provides even and accurate spraying so that users can save liquid and reduce operating costs.



Liquid Level Gauge



Flow Meter Module



Spray Tank Liquid Level Gauge

The liquid level gauge is used to obtain the amount of the liquid, making measurements more accurate than ever, which supports making alerts when the tank is empty.

02 Material Identification—Interchangeable Modules



Omnidirectional Digital Radar



Upward Radar



FPV Module



RTK Antenna Module



ESC Module



Flow Meter Module



Aerial-Electronics System



Spraying Module



RF Module

The modules mentioned above are compatible with both the T10 and T30.

02 Material Identification—Exclusive Modules



Power Distribution Module



10010S Motor



Cable Distribution Board Module



Delivery Pump



Liquid Level Gauge



Electromagnetic Exhaust Valves



Spray Tank Liquid Level Gauge



Rear Indicator Module

T10



Power Distribution Module



Cable Distribution Board Module



Liquid Level Gauge



10018 Motor



Plunger Pump



Electromagnetic Exhaust Valves

T30



02 Material Identification—Compatibility List

Motor

Material Name	Material Number	T16	T20	T10	T30
10015 Motor (Thickness)	BC.AG.AA000137.02	✓	×	×	×
10015 Motor (Thin)	BC.PR.AA000090.01	×	✓	×	×
10010S Motor	BC.AG.AA000336.01	×	×	✓	×
10018 Motor	BC.AG.AA000335.01	×	×	×	✓
Silicone Rubber Pad	YC.XJ.QT000196.04	✓	✓	✓	✓
Teflon Gasket	YC.SJ.WS002416.03	✓	✓	✓	✓
Aluminum Alloy Propeller Holder CCW	YC.JG.QX000240.01	×	✓	✓	✓
Aluminum Alloy Propeller Holder CW	YC.JG.QX000238.03	×	✓	✓	✓
33-inch Propeller CCW	YC.SJ.GG000060.01	✓	✓	✓	×
33-inch Propeller CW	YC.SJ.GG000061.01	✓	✓	✓	×
38-inch Propeller CCW	YC.JG.ZS000813.01	×	×	×	✓
38-inch Propeller CW	YC.JG.ZS000812.01	×	×	×	✓

Aircraft

Module	T16	T20	T10	T30
Aerial-Electronics System Module	x	x	✓	✓
Spraying Module	x	x	✓	✓
RF Module	x	x	✓	✓
RTK Module	x	x	✓	✓
Power Distribution Board	x	x	x	x
Cable Distribution Board	None	None	x	x
Radar	x	x	✓	✓
Delivery Pump	✓	✓	x	x
Electromagnetic Exhaust Valves	None	x	x	x
FPV Module	x	x	✓	✓
Liquid Level Gauge	x	x	x	x
Upward Radar	None	None	x	x
ESC	x	x	✓	✓
Remote Controller	x	x	✓	✓



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03 Repair Strategy

03 Repair Strategy

Product	Component	Dealers
Charger	4850 Power Adapter	Replace the whole module.
	Charger Main Board	Replace the whole module.
	LED Board Module	Replace the whole module.
	Power Module Adaptive Board	Replace the whole module.
	Structure Part	Repair all components.
Battery	Battery Cell Module	Not related
	Upper Shell Module (Including Battery Main Board)	Not related
	Appearance Part (battery level sticker, handle, bottom pad)	Replace the appearance part such as battery level sticker, handle, and bottom pad.
Remote Controller Main Part	Remote Controller	Repair all components (currently, the main board will be repaired)
Remote Controller Expansion Module	Remote Controller Expansion Module	Repair all components.
Aircraft	Aerial-Electronics System Module	Replace the whole module.
	Spraying System Module	Replace the whole module.
	RF Module	Replace the whole module.
	Distribution Board Module	Replace the whole module. (Repair the barometer.)
	FPV Module	Replace the whole module.
	Aircraft Arm	Replace the whole motor module.
		Replace the whole ESC module.
		Repair other components.
	Omnidirectional Digital Radar	Replace the whole module.
	Upward Radar	Replace the whole module.
	Flow Meter Module	Replace the whole module.
	Diaphragm Pump	Out of warranty: Repair all components. Under warranty: Replace the whole module.
	Power Hub Module	Replace the whole module.
	RTK Antenna Module	Replace the whole module.
	Landing Gear	Repair all components.
	Rear Indicator Module	Replace the whole module.
	Liquid Level Gauge	Replace the whole module.
	Spray Tank	Repair all components. (Replace the T10 spray tank one-way valve.)
	Electromagnetic Exhaust Valve Module	Replace the whole module.
	Nozzle & Spray Lance & Hose	Repair all components.
	Propellers CCW & CW	If one of the two propellers on one motor is damaged, perform damage assessment on the two propellers and deliver the two propellers.

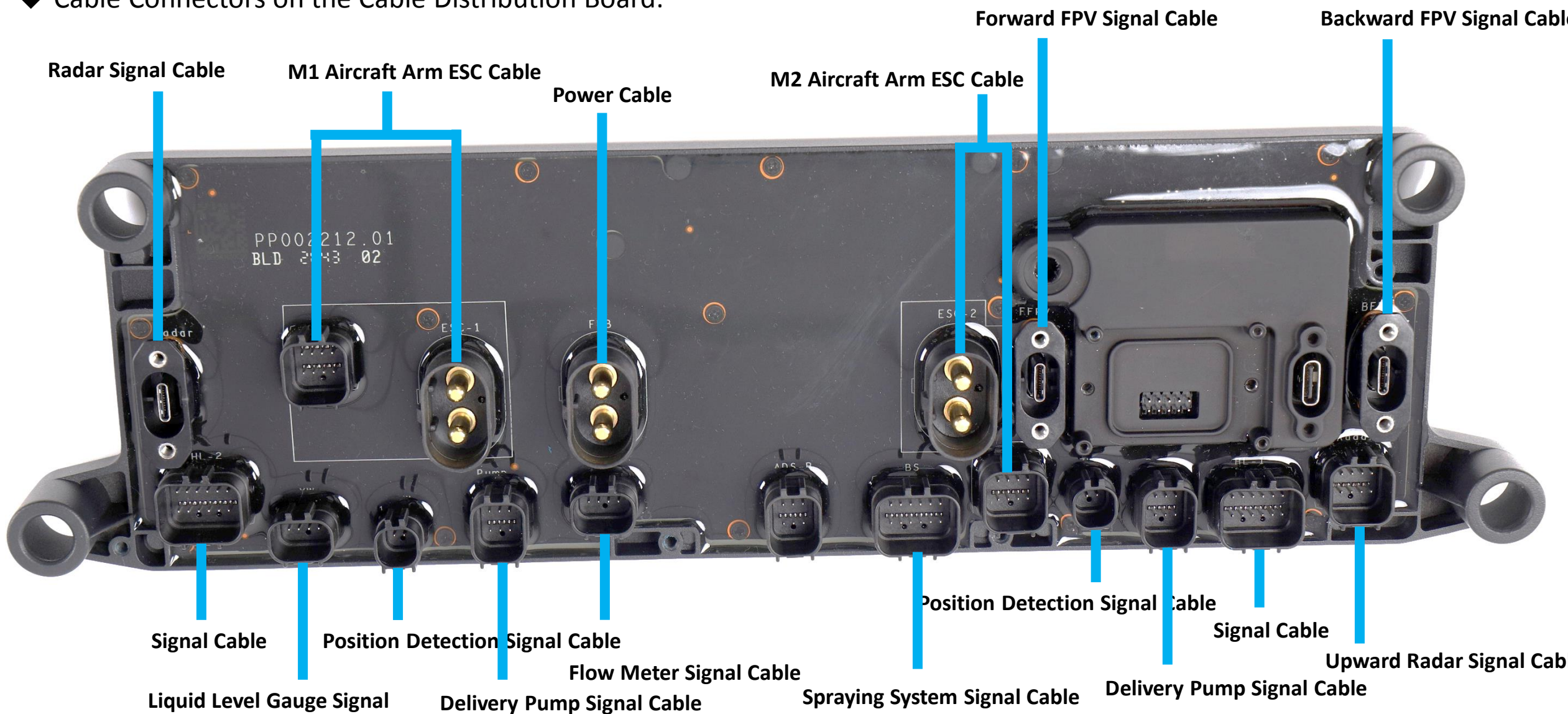


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04 Disassembly and Assembly Notes

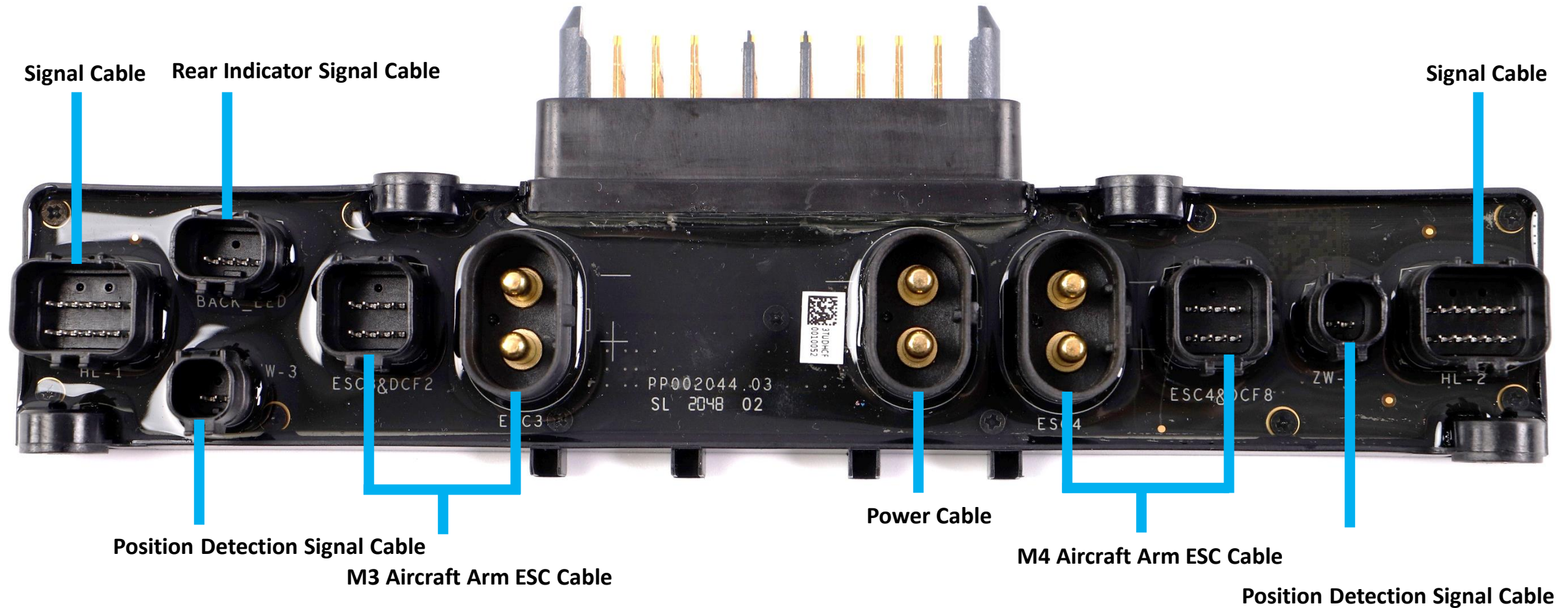
04 Disassembly and Assembly Notes

◆ Cable Connectors on the Cable Distribution Board:



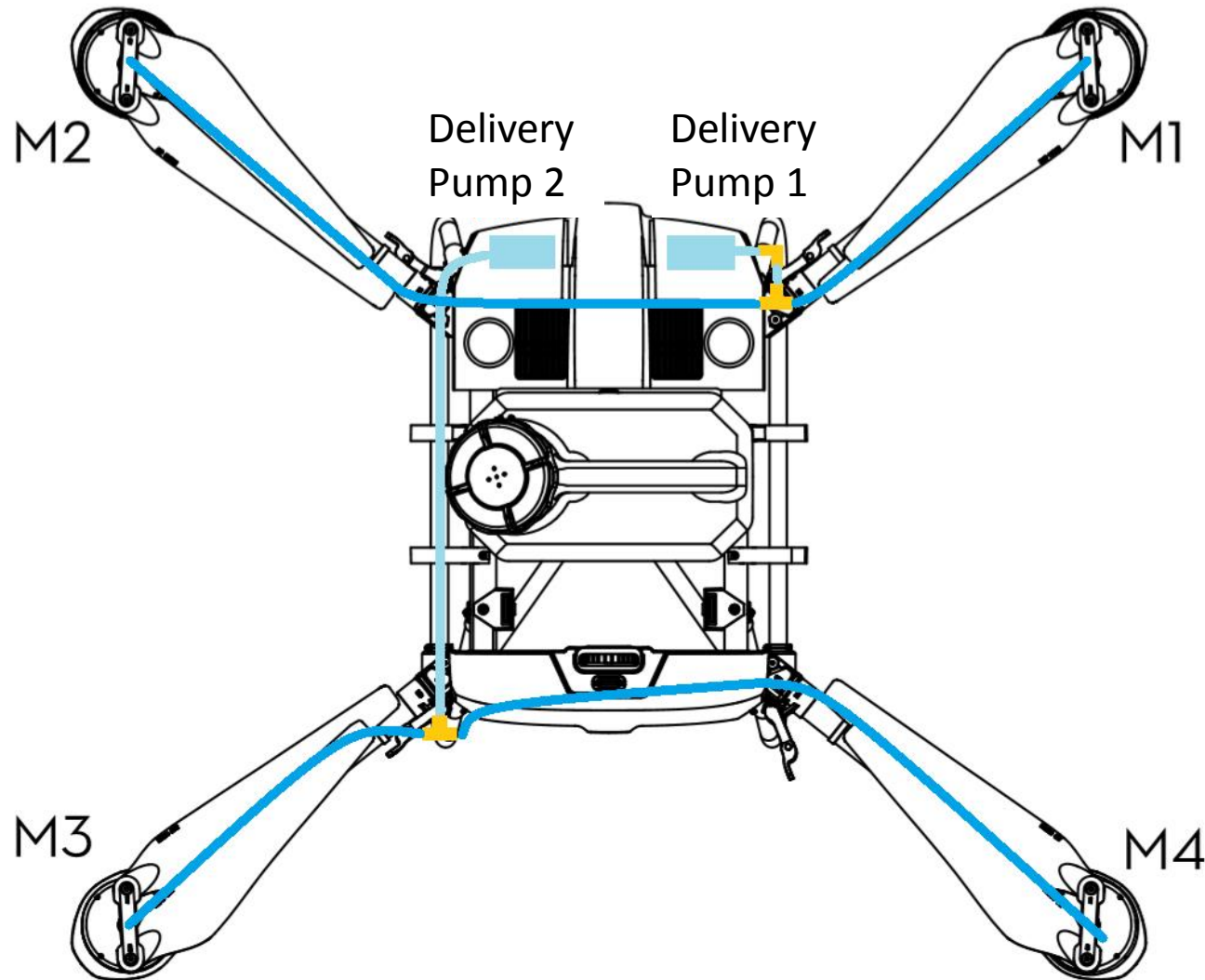
04 Disassembly and Assembly Notes

◆ Cable Connectors on the Power Distribution Board:



04 Disassembly and Assembly Notes

◆ Spraying System Hose Position:



Note :

1. Pay attention to the position of delivery pumps 1 and 2.
2. Pay attention to the connection method between the hoses and two delivery pumps.

04 Disassembly and Assembly Notices - Aircraft

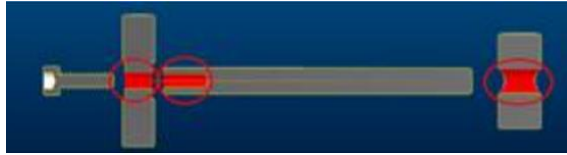


When replacing the propellers or propeller holder, the screw needs to be replaced and the screw glue needs to be applied.



When replacing the screw on the motor or replacing the motor, ensure that the screw is mounted in place. Clean off the screw glue and apply it again. Otherwise the motor may be loose and the aircraft vibrates greatly.

04 Disassembly and Assembly Notes - Aircraft

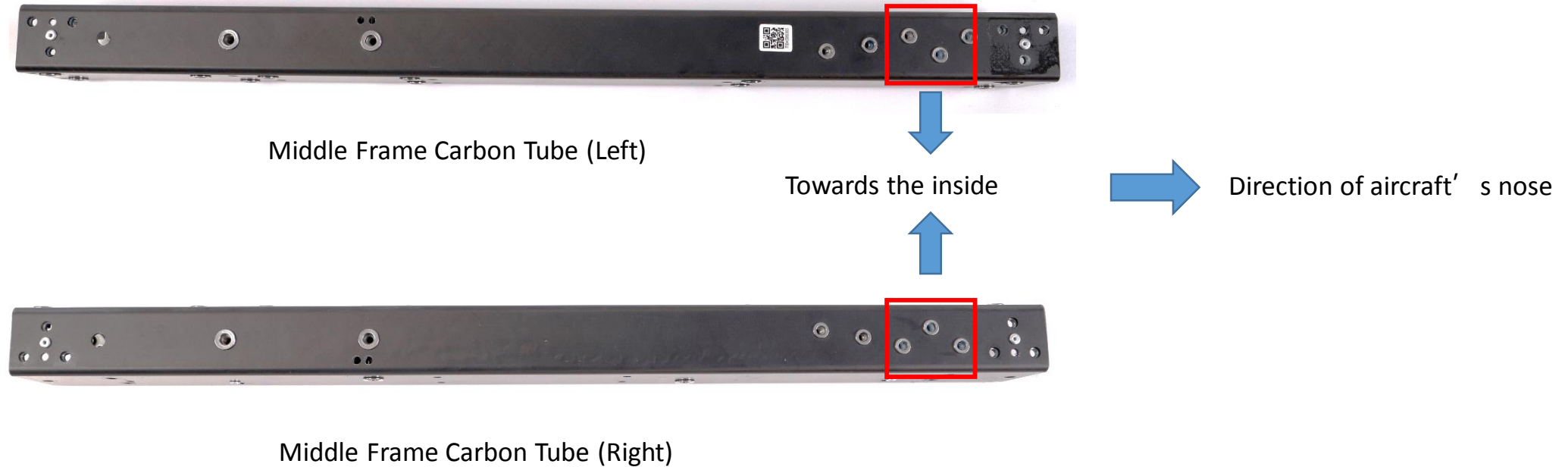


When replacing the aircraft arm lock, apply the FL.SC.H00193 Loctite 263 Loctite 263 Red Threadlocker.



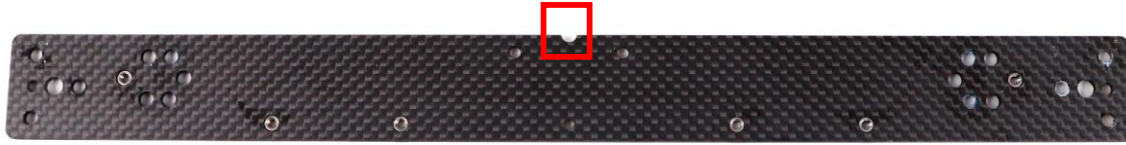
After the arm lock is replaced, fasten the lever as shown. The tension value of the arm lock needs to be tested (acceptable tension range: $80 \pm 5\text{N}$).

04 Disassembly and Assembly Notes - Aircraft



1. Distinguish between the direction of the left and right middle frame carbon tubes.
2. There is a triangle arrow mark on the front of the carbon tube near the front frame, and the arrow indicates the direction towards the inside of the middle frame.

04 Disassembly and Assembly Notes - Aircraft



Front Frame Upper Carbon Board



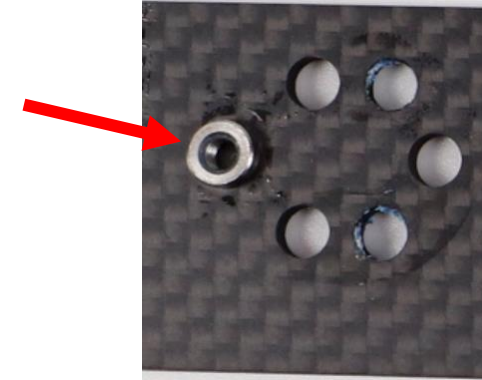
Front Frame Lower Carbon Board



Rear Frame Upper Carbon Board



Rear Frame Lower Carbon Board



1. The structure of the front and rear frame carbon boards is different.
2. There are slots on the front and rear frame upper carbon boards
3. The nut on the carbon board is raised. Make sure that the upper carbon board faces downward and the lower carbon board faces upward during assembly.

04 Disassembly and Assembly Notes - Aircraft



Landing Gear Fixing Base
(Front Left and Rear Right)



Landing Gear Fixing Base (Front
Right and Rear Left)

The landing gear fixing base needs to be mounted facing outwards.

The front left and rear right landing gear fixing bases are interchangeable.

The front right and rear left landing gear fixing bases are interchangeable.



Right Aircraft
Arm Connecting
Part



Left Aircraft Arm
Connecting Part

The M2 and M3 aircraft arm connecting parts are interchangeable, and the M1 and M4 aircraft arm connecting parts are interchangeable.

There are laser engraving marks on the upper and lower sides of the aircraft arm separately.

04 Disassembly and Assembly Notes - Aircraft



Aircraft Arm
Fixing Piece (Left)



Aircraft Arm
Fixing Piece
(Right)

The M2 and M3 aircraft arm fixing pieces are interchangeable, and the M1 and M4 aircraft arm fixing pieces are interchangeable.



Pay attention to the installation direction of the landing gear, and the radar signal cable needs to be attached to the left side.

04 Disassembly and Assembly Notes – Remote Controller

Removing the lower shell

There is a buckle between the upper and lower shells.

There are buckles between the upper and lower shells, which are very firm after assembly.

When removing the lower shell, first unscrew all the screws, hold the lower shell with your hand, and remove it forcefully.



Removing the front cover's decorative piece

When removing the components inside the front shell, remove the front cover's decorative piece first. Remove the two screws first, and then remove the front cover's decorative piece by using the crowbar to pry it off.



04 Disassembly and Assembly Notes – Remote Controller

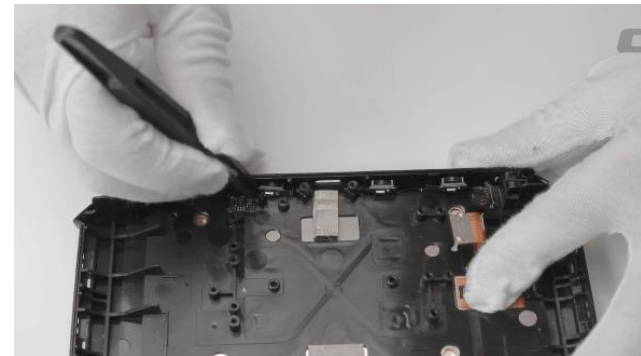
Removing the antenna sleeve

When removing the antenna sleeve or replacing the image transmission antenna, keep in mind that glue was applied during assembly to make the product very firm, so the heat gun needs to first be used to blow on the antenna sleeve first (minimum temperature and minimum air volume). When the sleeve becomes misshapen, shake the sleeve to the left and right, and then remove and discard it. Do not damage the antenna base with the heat gun.



Removing the Mic board

There are four fixing pieces (buckles) on the Mic board. Observe the mounting position, and then use the tweezers to eject them.

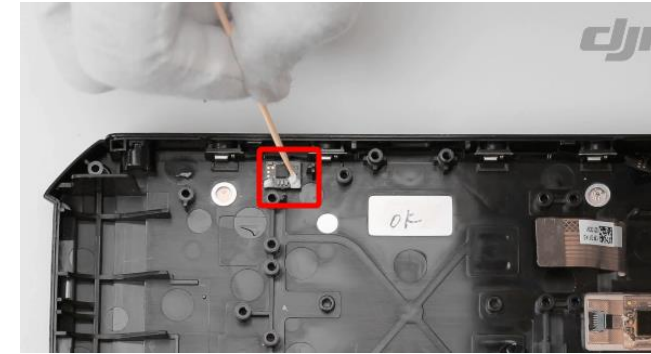


04 Disassembly and Assembly Notes – Remote Controller

Mounting the Mic board

When the Mic board is mounted, a hot-melt process is used. After repair, white glue needs to be applied.

Note: White Glue GD-883D



Mounting left and right soft rubber

1. Apply the Kafuter instant adhesive before mounting the left and right soft rubbers.
2. Mounting the 5D button. When the 5D button is replaced, the Kafuter instant adhesive needs to be applied. Then apply a little amount of hot melt glue to the button on the button board.

Note: K-4323N_Instant Adhesive or 3M PR1500



04 Disassembly and Assembly Notes – Remote Controller

Consumption Materials

Left and right soft rubber

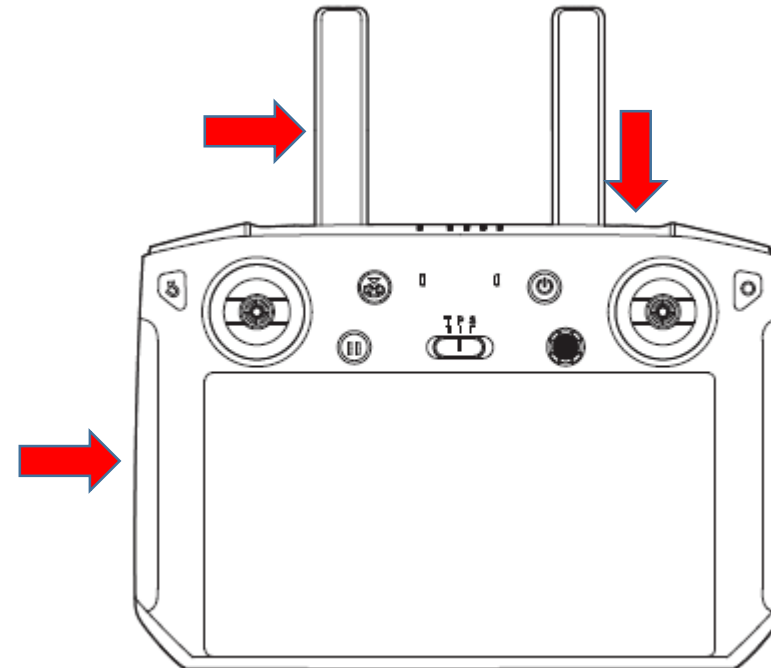
Replace with new rubbers after disassembling the remote controller.

Antenna sleeve

Replace with a new antenna sleeve after replacing the antenna.

Front cover's decorative piece

Replace it when disassembling the components of the front cover.

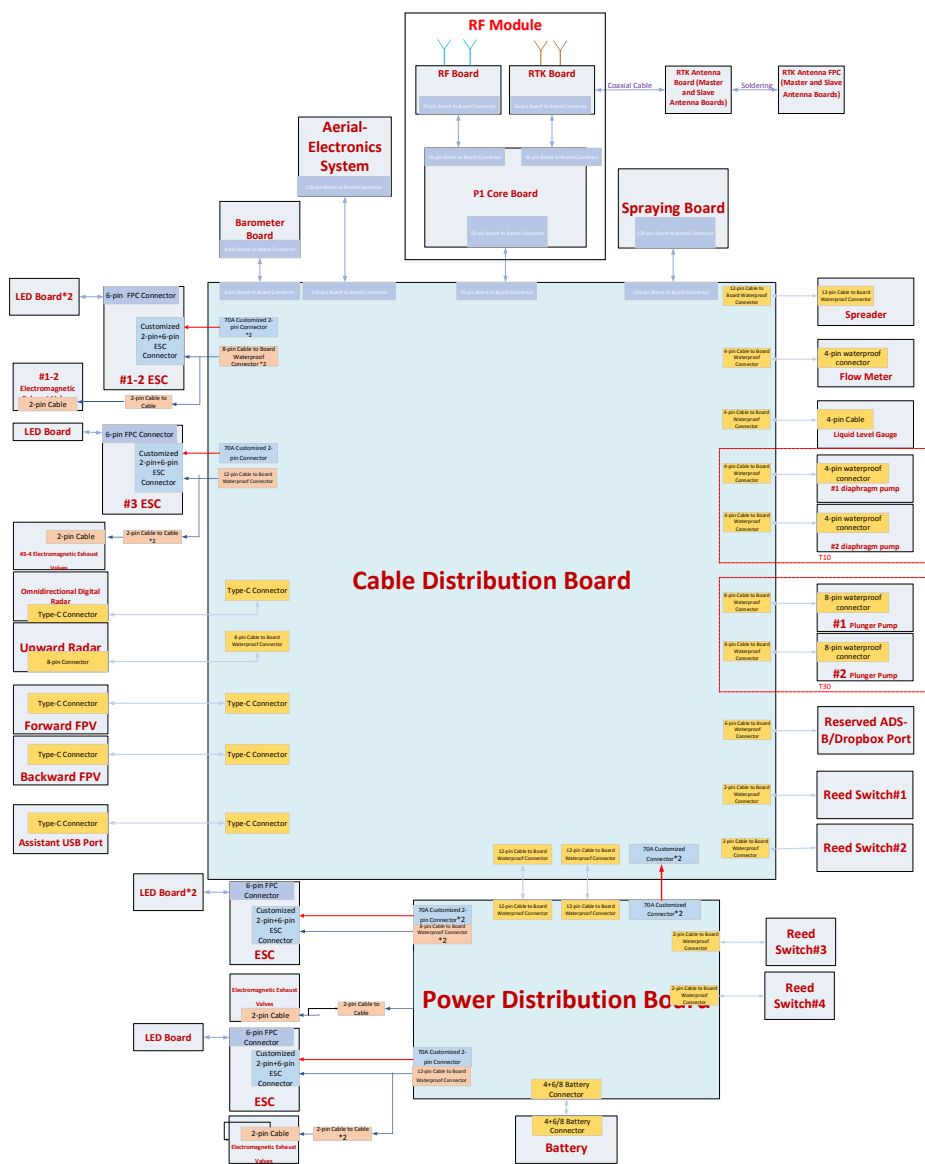




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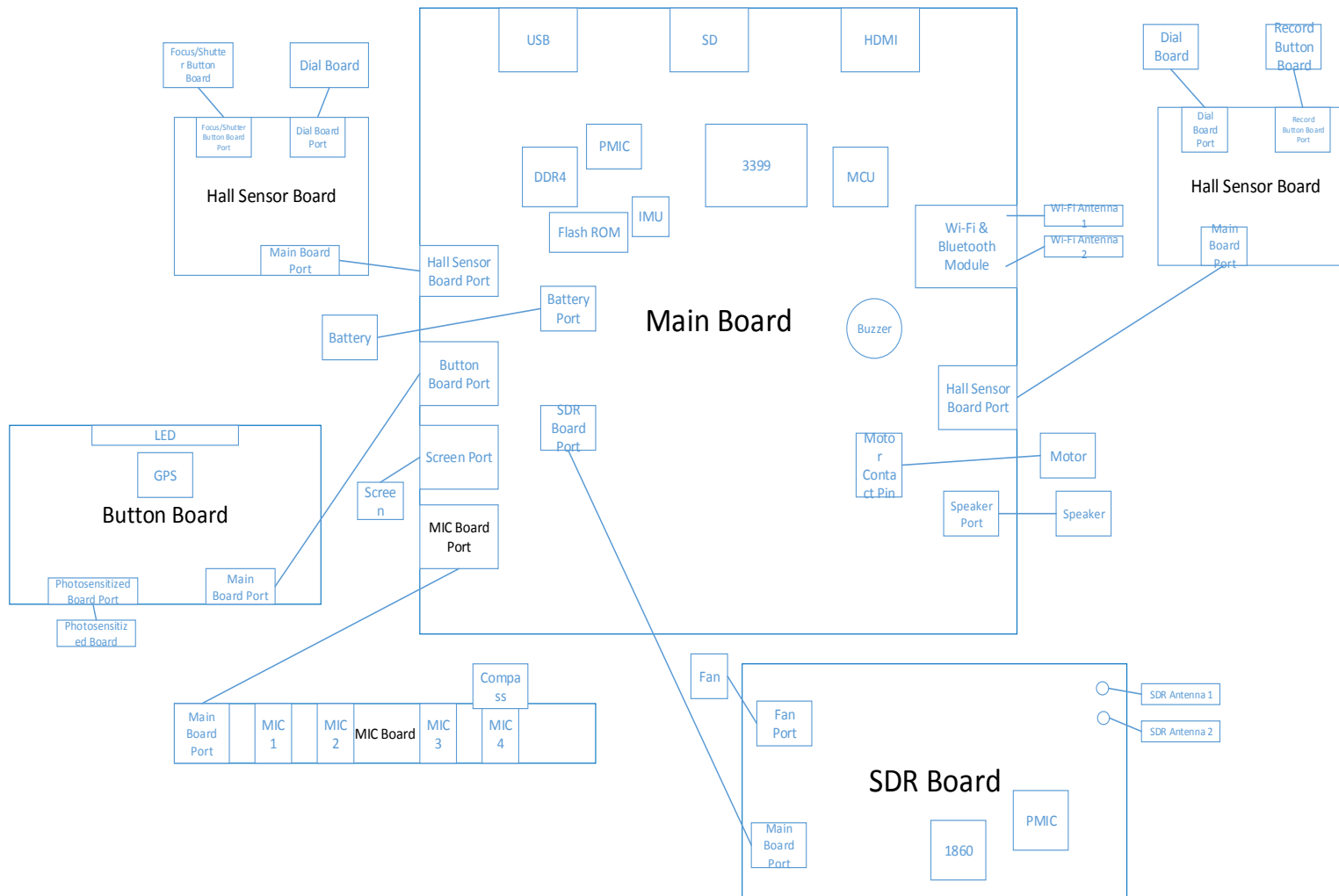
05 Principles Diagram and Boards Introduction

05 Principles Diagram - Aircraft



System	Module/Board	Function
Aerial-Electronics System	Aerial-Electronics System Module	The Aerial-Electronics Board is the core module controlling the flight controller. It integrates the flight algorithm, image transmission, and flight controlling functions into one module. The compass is inside the module.
	Motor	The motors work with propellers to blow wind downwards and make the aircraft fly upwards. It also helps with spraying. The six motors are interchangeable.
Propulsion System	ESC Module	The ESC board converts the throttle information to the motor's rotation speed information, controls the motor rotation, and sends the motor information to the aerial-electronics board.
	LED	The LED board indicates the aircraft's current working condition. Users can know the aircraft's current working condition by the LED's color.
	Power Hub Board	The power transfer module that distributes the power to the six motors, and is the transfer station for the battery and power supply.
	Cable Distribution Board	Connects the cables and modules and includes a barometer.
	Barometer	Receives the current barometric pressure signal, sends the barometric pressure information to the aerial-electronics board, and detects the aircraft's current altitude information.
Spraying System	Spraying System Module	The control system of the aircraft's core spraying function. It controls the pump and spraying volume
	Liquid Level Gauge	For T30, it is a continuous liquid level gauge to measure the current liquid amount in the tank to notify the customer whether to add liquid. For T10, similar with T20.
	2-Channel Flow Meter	It is essential for high-precision spraying. It detects the pump's spraying speed. Therefore, the pump's spraying accuracy can be controlled.
	Delivery Pump	The water pump is the exerciser of the aircraft spraying system, which sprays pesticides out by rotating. For T30, the plunger pump is used, and diaphragm pump for T10.
	Electromagnetic Exhaust Valves	Adjust the air in the water pipe, and can independently turn on or off any nozzle to spray.
Positioning System	RF Module (Baseband Board, RF Board, RTK/GPS Board)	As a flight controller system to provide the aircraft position, direction, and speed of the aircraft to ensure the flight stability, and the efficiency and accuracy of the operation. It can track GPS L1/L2, Glonass F1/F2, BDS B1/B2, and Galileo E1/E5b frequency bands at the same time. It has positioning and orientation functions. The GPS module uses the main antenna of RTK module to receive GPS L1, and Glonass F1 frequency band signals.
	Active Antenna Board	Amplify and adjust the weak GPS signal so that the RTK/GPS board can get better satellite signals
Radar System	Omnidirectional Digital Radar	A sensor for obstacle avoidance and navigation of aircraft. Responsible for investigating the external environment to prevent the aircraft from hurting people and crash.
	Upward Radar	One of the sensors to detect the environment above the aircraft to prevent collisions with obstacles while ascending.
Image Transmission system	Forward and Backward FPV Cameras	Converts optical signals into digital signals, and transmits them to customers through image transmission. The FPV camera contains LEDs to provide night lighting.
	RF Module (Baseband Board, RF Board, RTK/GPS Board)	Includes the baseband board and RF board, processing the image transmission signal and connecting the aircraft and remote controller.
Battery System	Battery	Batteries are the source of energy for flight and spraying. It is an energy storage device.

05 Principles Diagram – Remote Controller



Chip	Function
3399	Main functional chip
MCU	Micro control unit, 1549 chip
PMIC	Power management chip
DDR4	Fourth-generation memory chip; temporary storage unit for commands and data.
Flash ROM	Store program and data
IMU	Inertial measurement unit; including accelerometer and gyroscope, related to gravity sensing
1860	Wireless image transmission bandwidth chip; used for digital signal processing, encoding, and digital signal modulation.

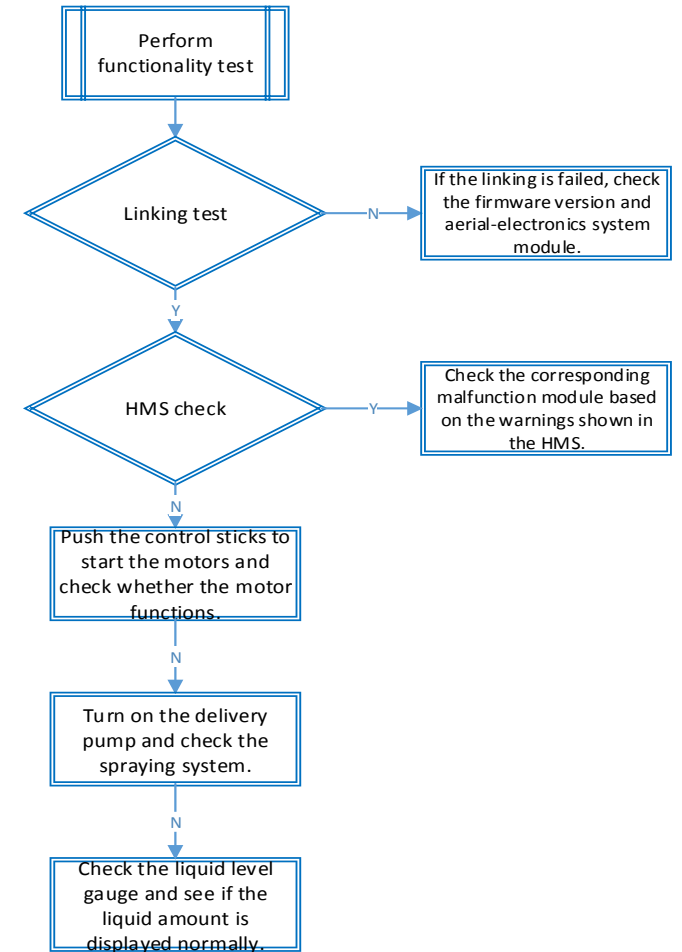
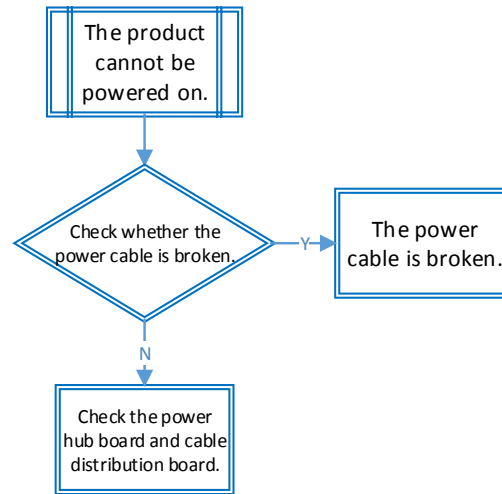
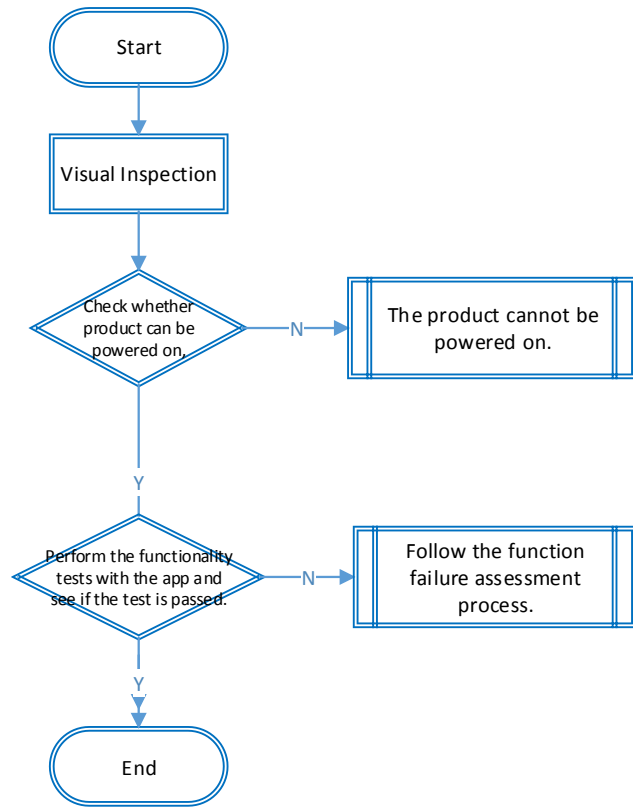


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06 Malfunction Assessment and Analysis

06 Malfunction Assessment and Analysis—Common Issues

Using HMS in the app for damage assessment.





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FAQ



Thank You!